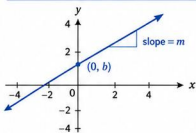


Precalculus - Important Functions

Key function families, graphs, and formulas

1 Linear Function

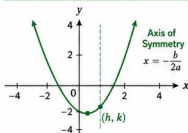
$$y = mx + b$$



- Constant rate of change.
- Slope (m) determines steepness and direction.
- y -intercept is $(0, b)$.
- Domain and range: all real numbers.

2 Quadratic Function

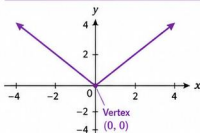
$$y = ax^2 + bx + c \quad (a \neq 0)$$



- Parabola shape (opens up if $a > 0$, down if $a < 0$).
- Vertex is (h, k) , where $h = -\frac{b}{2a}$.
- Axis of symmetry: $x = -\frac{b}{2a}$.
- Domain: all real numbers; Range depends on a .

3 Absolute Value Function

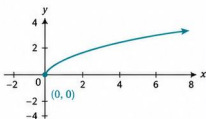
$$y = |x|$$



- V-shape graph.
- Vertex at $(0, 0)$.
- Symmetric about the y -axis.
- Domain and range: all real numbers, $y \geq 0$.

4 Square Root Function

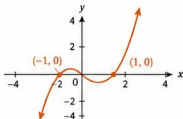
$$y = \sqrt{x}$$



- Starts at $(0, 0)$ and exists for $x \geq 0$.
- Increasing and concave down.
- Domain: $[0, \infty)$; Range: $[0, \infty)$.

5 Polynomial Function

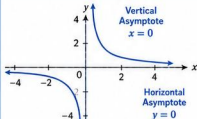
$$y = x^3 - x$$



- Example is degree 3 (cubic).
- End behavior: down on left, up on right.
- Has turning points (local max and min).
- Domain and range: all real numbers.

6 Rational Function

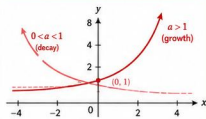
$$y = \frac{1}{x}$$



- Asymptotes: $x = 0$ (vertical), $y = 0$ (horizontal).
- Not defined at $x = 0$.
- Domain: $(-\infty, 0) \cup (0, \infty)$; Range: $(-\infty, 0) \cup (0, \infty)$.

7 Exponential Function

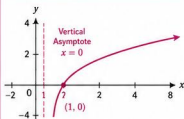
$$y = a^x \quad (a > 0, a \neq 1)$$



- Rapid growth if $a > 1$; decay if $0 < a < 1$.
- Passes through $(0, 1)$.
- Horizontal asymptote: $y = 0$.
- Domain: all real numbers; Range: $(0, \infty)$.

8 Logarithmic Function

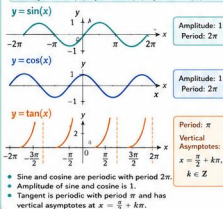
$$y = \log(x) \quad (\text{common log})$$



- Inverse of exponential function.
- Vertical asymptote at $x = 0$.
- Defined for $x > 0$ only.
- Domain: $(0, \infty)$; Range: all real numbers.

9 Trigonometric Functions

$$y = \sin(x), y = \cos(x), y = \tan(x)$$



- Sine and cosine are periodic with period 2π .
- Amplitude of sine and cosine is 1.
- Tangent is periodic with period π and has vertical asymptotes at $x = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$.

Quick Reminders



Domain

All possible input values (x) for which the function is defined.



Range

All possible output values (y) the function can produce.



Intercepts

x -intercepts: set $y = 0$.
 y -intercept: set $x = 0$.
Points where the graph crosses the axes.



Symmetry

Even functions are symmetric about the y -axis.
Odd functions are symmetric about the origin.



Transformations

Shifts, stretches, compressions, and reflections change the graph's position or shape.



Asymptotes

Lines the graph approaches but does not cross (vertical, horizontal, or slant).